

• Notes for lecture 1

Recall that

$$L^1_{loc}(\mathbb{R}^d) := \left\{ f \text{ Lebesgue measurable in } \mathbb{R}^d : \int_K |f(y)| dy < \infty \right. \\ \left. \text{for every } K \subseteq \mathbb{R}^d \text{ compact} \right\}.$$

For $f \in L^1_{loc}(\mathbb{R}^d)$, define its centred maximal function $M_c(f)$ by

$$M_c(f)(x) := \sup_{r>0} \frac{1}{|B(x,r)|} \int_{B(x,r)} |f(y)| dy \quad (x \in \mathbb{R}^d)$$

and its non-centred maximal function $M(f)$ by

$$M(f)(x) := \sup_{\substack{B \subseteq \mathbb{R}^d \\ \text{open ball:} \\ x \in B}} \frac{1}{|B|} \int_B |f(y)| dy \quad (x \in \mathbb{R}^d).$$

Here, for a Lebesgue measurable set A in $\mathbb{R}^d$, $|A|$ stands for its Lebesgue measure. Also, for $x \in \mathbb{R}^d$ and $r > 0$, $B(x,r)$ denotes the open ball in $\mathbb{R}^d$ with centre x and radius r ; $B(x,r) := \{y \in \mathbb{R}^d : |x-y| < r\}$. Note that $|B(x,r)| = v_d \cdot r^d$, where v_d is the volume of the unit ball in $\mathbb{R}^d$, namely $v_d = |B(0,1)| = \pi^{d/2} / \Gamma(\frac{d}{2} + 1)$.

Exercise. Show that for $f \in L^1_{loc}(\mathbb{R}^d)$ one has $M_c(f) \leq M(f) \leq 2^d M_c(f)$ a.e. on $\mathbb{R}^d$.

Remarks. 1) If $f \in L^\infty(\mathbb{R}^d)$ then $M(f) \in L^\infty(\mathbb{R}^d)$ with

$$\|M(f)\|_{L^\infty(\mathbb{R}^d)} \leq \|f\|_{L^\infty(\mathbb{R}^d)}.$$

2) If $f \in L^1(\mathbb{R}^d)$ then $M(f) \notin L^1(\mathbb{R}^d)$ unless f equals zero a.e. on $\mathbb{R}^d$.

Indeed, if $f \in L^1(\mathbb{R}^d)$ is not zero a.e. on $\mathbb{R}^d$ then there exists an $R > 0$ such that $\int_{B(0,R)} |f(y)| dy > 0$. Hence, for $|x| > R$ we have

$$\begin{aligned} M(f)(x) &\geq \frac{1}{|B(0, R+|x|)|} \int_{B(0, R+|x|)} |f(y)| dy \\ &\geq \frac{1}{|B(0, R+|x|)|} \int_{B(0, R)} |f(y)| dy \quad (\text{as } B(0, R) \subseteq B(0, R+|x|)) \\ &= \frac{1}{V_d \cdot (R+|x|)^d} \int_{B(0, R)} |f(y)| dy \\ &\geq \frac{1}{V_d \cdot (2|x|)^d} \int_{B(0, R)} |f(y)| dy \quad (\text{as } R+|x| \leq 2|x|) \\ &= C_{d,f} \cdot \frac{1}{|x|^d}, \quad \text{where } C_{d,f} = (V_d \cdot 2^d)^{-1} \int_{B(0, R)} |f(y)| dy > 0. \end{aligned}$$

Hence,

$$\int_{\mathbb{R}^d} M(f)(x) dx \geq \int_{|x| > R} M(f)(x) dx \geq C_{d,f} \int_{|x| > R} |x|^{-d} dx = \infty.$$

Even though that M is not bounded on $L^1(\mathbb{R}^d)$, we have the following 'weak-type' inequality.

Theorem 1. For every $f \in L^1(\mathbb{R}^d)$ one has

$$|\{x \in \mathbb{R}^d : M(f)(x) > a\}| \leq \frac{3^d}{a} \int_{\mathbb{R}^d} |f(x)| dx$$

for all $a > 0$.

Proof. Let $f \in L^1(\mathbb{R}^d)$ and $a > 0$ be fixed. Write

$$E_a := \{x \in \mathbb{R}^d : M(f)(x) > a\}$$

and assume, without loss of generality, that $E_a \neq \emptyset$.

Exercise. Show that E_a is open

By the inner regularity of the Lebesgue measure, we have

$$|E_a| = \sup_{\substack{K \subseteq E_a \\ \text{compact}}} |K|.$$

Hence, it suffices to prove that

$$(*) \quad |K| \leq \frac{3^d}{a} \int_{\mathbb{R}^d} |f(x)| dx \quad \text{for all } K \subseteq E_a \text{ compact.}$$

Let $K \subseteq E_a$ be a fixed, non-empty compact set.

Let $x \in K$. Then $x \in E_a$ and so, by the definition of E_a , there exists an open ball B_x containing x such that

$$(I) \quad |B_x| < \frac{1}{a} \int_{B_x} |f(y)| dy.$$

Hence, $K \subseteq \bigcup_{x \in K} B_x$ and since K is compact, there exist $x_1, \dots, x_N \in K$ such that

$$\bullet K \subseteq \bigcup_{\ell=1}^N B_{x_\ell}$$

$$\bullet B_{x_\ell} \text{ satisfies (I) for } \ell \in \{1, \dots, N\}.$$

To complete the proof of $(*)$, we shall employ the following Vitali-type covering lemma.

Lemma 2. For every finite collection $\{B_\ell\}_{\ell=1}^N$ of (open) balls there exists a sub-collection $\{B_{\ell_j}\}_{j=1}^M$ of $\{B_\ell\}_{\ell=1}^N$ such that:

$$\bullet \text{ the balls in } \{B_{\ell_j}\}_{j=1}^M \text{ are mutually disjoint and}$$

$$\bullet \left| \bigcup_{j=1}^M B_{\ell_j} \right| \leq 3^d \left| \bigcup_{\ell=1}^N B_\ell \right|.$$

Proof. Consider a relabeling $\{B'_k\}_{k=1}^N$ of $\{B_\ell\}_{\ell=1}^N$ that is such that

$$|B'_1| \geq |B'_2| \geq \dots \geq |B'_N|.$$

Choose $B_{\ell_1} := B'_1$. Having chosen $B_{\ell_1}, \dots, B_{\ell_j}$ for some $j \in \{1, \dots, N-1\}$, consider the set of indices

$$A_j := \{k \in \{\ell_j+1, \dots, N\} : B'_k \cap \left(\bigcup_{\ell=1}^j B_{\ell_\ell} \right) = \emptyset\}$$

If $A_j \neq \emptyset$, choose $\ell_{j+1} := \min A_j$ and $B_{\ell_{j+1}} := B'_{\min A_j}$. Otherwise, the process terminates. As the original collection is finite the process terminates in M steps, $M \leq N$.

By the selection process, the balls $\{B_{\ell_j}\}_{j=1}^M$ are mutually disjoint. It remains to show that

$$\left| \bigcup_{k=1}^M B'_k \right| \leq 3^d \left| \bigcup_{\ell=1}^N B_{\ell_\ell} \right|.$$

To this end, observe that for each $k \in \{1, \dots, N\}$ B'_k either belongs to $\{B_{\ell_j}\}_{j=1}^M$ or not. In the latter case, there exists a $j_0 \in \{1, \dots, M\}$ such that $B'_k \cap B_{\ell_{j_0}} \neq \emptyset$ and $|B'_k| \leq |B_{\ell_{j_0}}|$.

Exercise. Show that if $B(x, r)$ and $B(x', r')$ are such that :

- $B(x, r) \cap B(x', r') \neq \emptyset$ and
- $|B(x, r)| \leq |B(x', r')|$ (equivalently, $r \leq r'$)

then $B(x, r) \subseteq 3B(x', r')$, where $3B(x', r') := B(x', 3r')$.

It follows from the exercise above that

$$\bigcup_{\ell=1}^N B_{\ell} = \bigcup_{k=1}^N B'_k \subseteq \bigcup_{j=1}^M 3B_{\ell_j}.$$

Hence,

$$\begin{aligned} \left| \bigcup_{\ell=1}^N B_{\ell} \right| &\leq \left| \bigcup_{j=1}^M 3B_{\ell_j} \right| \leq \sum_{j=1}^M |3B_{\ell_j}| \\ &= \sum_{j=1}^M 3^d |B_{\ell_j}| \\ &= 3^d \left| \bigcup_{j=1}^M B_{\ell_j} \right| \quad (\text{as the balls in } \{B_{\ell_j}\}_{j=1}^M \text{ are mutually disjoint}) \end{aligned}$$

and so, the proof of the lemma is complete. $\blacksquare$

If $K \subseteq \bigcup_{\ell=1}^N B_{x_{\ell}}$ with $B_{x_{\ell}}$ satisfying (I), then Lemma 2 implies that there exists a sub-collection $\{B_{x_{\ell_j}}\}_{j=1}^M$ of $\{B_{x_{\ell}}\}_{\ell=1}^N$ consisting of mutually disjoint balls such that $\left| \bigcup_{\ell=1}^N B_{x_{\ell}} \right| \leq 3^d \left| \bigcup_{j=1}^M B_{x_{\ell_j}} \right|$. Hence,

$$\begin{aligned} |K| &\leq \left| \bigcup_{\ell=1}^N B_{x_{\ell}} \right| \leq 3^d \left| \bigcup_{j=1}^M B_{x_{\ell_j}} \right| = 3^d \sum_{j=1}^M |B_{x_{\ell_j}}| \\ &< 3^d \sum_{j=1}^M \frac{1}{\alpha} \int_{B_{x_{\ell_j}}} |f(y)| dy \quad (\text{by (I)}) \\ &= \frac{3^d}{\alpha} \int_{\bigcup_{j=1}^M B_{x_{\ell_j}}} |f(y)| dy \leq \frac{3^d}{\alpha} \int_{\mathbb{R}^d} |f(y)| dy. \end{aligned}$$

This completes the proof of Theorem 1. $\blacksquare$

The material of this lecture is based on

- Grafakos, Loukas. Classical Fourier Analysis. Third Edition. Graduate Texts in Mathematics, 249. Springer. New York, 2014, xviii + 638 pp.